OJAS GRAMOPADHYE

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Education

New York University, New York — Aug '24 - May '26
Master of Science, Computer Science — *GPA : 3.89/4*
Specialization : *AI & ML, Natural Language Processing*
Courses : *Artificial Intelligence, Deep Learning, Robot Perception, Big Data, Machine Learning, Design & Analysis of Algorithms, Information Visualization*

Indian Institute of Technology Bombay, India — May '19 - Jul '23
Bachelor of Technology, Computer Science and Engineering — *GPA : 7.74/10*
Courses : *AI & ML, Speech Recognition, Image Processing, Database Systems, Discrete Mathematics, Data Structures & Algorithms, Software Systems*

Key Projects

Modulus : Agentic AI Assistant — Jun '25 - Ongoing
Independent Project — **New York University**

- Building an agentic assistant using the Model Context Protocol (MCP) and LangGraph to manage multi-threaded conversations, dynamic tool routing, and stateful LLM execution with real-time streaming
- Implementing vector-based memory using ChromaDB & MongoDB – backed persistence for short and long-term context tracking, with tool integration, a scalable FastAPI backend, and a React frontend

Flowsight : NYC Traffic Collision Analysis and Prediction — Jan '25 - May '25
Course Project | Big Data — **New York University**

- Developed a real-time big data pipeline using Apache Kafka and MongoDB to ingest, validate, and store live NYC collision and traffic speed data; implemented exponential backoff and retry mechanisms for fault-tolerance
- Engineered a Spark-based ML pipeline for crash-risk scoring using spatially matched traffic and collision data, training a Linear Regression model to identify high-risk locations with real-time dashboard visualization using React and Flask

Red Plag : Plagiarism Checker — Sept '20 - Dec '20
Course Project | Software Systems — **IIT Bombay**

- Developed a plagiarism detection system using language-specific tokenizers (C++, Java, Python) and winnowing of hashed k-grams for similarity computation, with a full-stack Angular–Django web platform for graphical visualization, database management (PostgreSQL), and secure access via JWT authentication.

Visual Perception - Lunar Autonomy Challenge — Jun '25 - Ongoing
Team Self Drive, NYU — **New York University**

- Working on NASA's Lunar Autonomy Challenge as part of NYU's Self Drive team, leveraging ORB-SLAM3 for visual SLAM and experimenting with FoundationStereo for robust depth estimation in challenging lunar environments

Autonomous Mobility Tool for Visually Impaired — Jan '25 - Jun '25
Team Sixth Sense, NYU — **New York University**

- Developed a vision-language Scene Description system for aiding the visually impaired by leveraging space-separated scene parsing, and methods like few-shot prompting, Chain of Thought reasoning, and iterative verification for LLMs

Work Experience

Detection and Mitigation of Hallucination in Generative Models | arχiv — May '23 - May '24
Research Assistant — **IBM Research | IIT Bombay**

- Researched hallucination mitigation in generative language models by creating a Subjective Medical QA dataset and developing a differential diagnosis system inspired by real medical practices; published in EMNLP Findings 2024.
- Achieved up to 90% accuracy by implementing a Forward-Backward Chain of Thought method and fine-tuning LLaMA-2-Chat (7B and 70B) models for reward generation with Hugging Face and Accelerate

Japanese OCR Pipeline — May '22 - Jul '22
Data Science Intern — **Daikin**

- Developed an OCR system using a MobileNetV3 + DB-based text detector on a dataset of 3000 proprietary company documents in Japanese, and achieved a Levenshtein Distance of 0.91 for character recognition

Locust Occurrence Modeling and Prediction — May '21 - Jul '21
Data Science Associate — **Dtime.ai**

- Predicted locust outbreaks in East Africa by engineering geospatial features and training a Bi-directional ConvLSTM U-Net for pixel-level classification on timestamped $10km^2$ grids, using SMOTE to handle data skew

Skills

Focus	Natural Language Processing (Modeling, Reasoning & Alignment), Agentic AI, Real-Time Data Pipelines
Frameworks	PyTorch, TensorFlow, HuggingFace, LangGraph, LangChain, OpenCV, DeepSpeed, Accelerate
Methods	LLMs, SLAM & Visual Perception, AI Agents, MLflow, Spark, Dask, Pandas, SQL, MongoDB, PostgreSQL, ChromaDB
Development	Python, C++, JavaCcript, Docker, Node.js, Angular, Django, React.js, d3.js